

**“Automated Application Deployment Using
GitHub and Docker”**

A SYNOPSIS SUBMITTED TO

THE NATIONAL INSTITUTE OF ENGINEERING, MYSURU

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in partial fulfillment for the award of degree of

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Introduction

DevOps is a modern software development approach that combines development and operations to improve software delivery speed, reliability, and automation. Docker provides containerization technology, while GitHub offers version control and collaboration features.

This project demonstrates the deployment of a web application using GitHub and Docker. The application is developed using Flask and integrated with Docker for containerized deployment. GitHub is used to manage source code and track project changes.

The project automates application deployment and ensures portability across different environments.

Objectives

- To understand DevOps workflow
- To implement version control using GitHub
- To containerize applications using Docker
- To deploy applications efficiently
- To improve scalability and portability
- To automate deployment process

Problem Statement

Traditional application deployment methods are time-consuming and environment-dependent. Developers often face issues related to dependency management and inconsistent deployment environments.

This project solves these issues by using:

- GitHub for source code management
- Docker for containerization
- Automated deployment workflow

Technologies Used

Technology	Purpose
Python Flask	Backend Web Application
HTML/CSS	Frontend UI
Git	Version Control
GitHub	Source Code Hosting
Docker	Containerization
Linux/Ubuntu	Deployment Environment

System Requirements

Hardware Requirements

- 4 GB RAM
- Intel i3 or above
- Internet Connection

Software Requirements

- Ubuntu/Linux
- Docker
- Git
- Python
- VS Code

Modules

1. Frontend Module

- User interface created using HTML and CSS

2. Backend Module

- Flask application handles routing and deployment

3. GitHub Integration Module

- Stores project source code
- Tracks changes using Git

4. Docker Deployment Module

- Builds Docker image
- Runs application in container

Advantages

- Easy deployment
- Faster software delivery
- Portable application environment
- Improved scalability
- Better version control
- Simplified maintenance

Applications

- Cloud deployment
- CI/CD pipelines
- Web application hosting
- DevOps automation
- Microservices architecture

Future Enhancements

- Jenkins CI/CD integration
- Kubernetes deployment
- AWS cloud hosting
- Monitoring using Grafana and Prometheus
- Automated testing pipeline

Conclusion

The project successfully demonstrates automated application deployment using GitHub and Docker. The integration of version control and containerization improves deployment efficiency, consistency, and scalability. This project provides practical exposure to modern DevOps practices and deployment automation.